SET 4: PRODUCT INVENTORY MANAGEMENT

Name	Mohamed Nazeer S
Register No.	192424084
Course Code & Name	DSA0601 – DATA HANDLING AND VISUALIZATION

DATASET

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

QUESTION 1 – BAR CHART: PRODUCT INVENTORY QUANTITY

AIM

To create a bar chart in RStudio showing the available quantity of each product.

ALGORITHM

1. Create product and quantity vectors.

2. Use barplot() to generate the chart.

3. Add axis labels and title.

4. Display the chart.

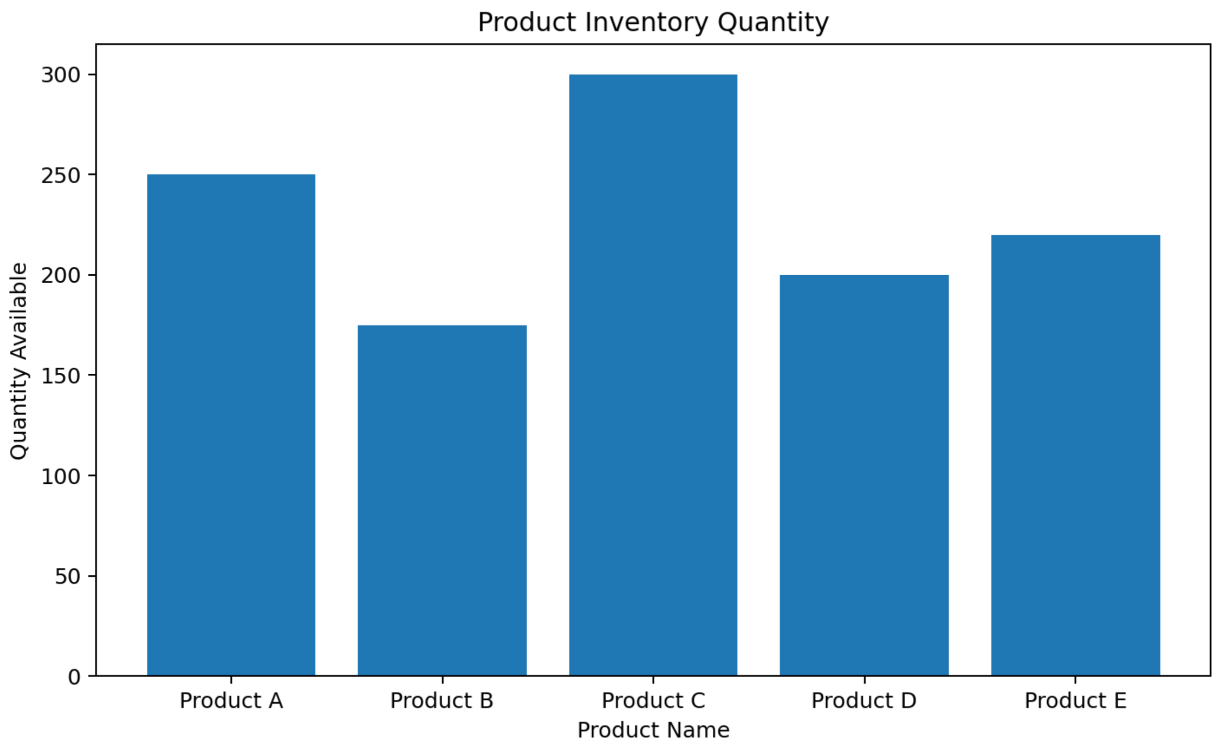
PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

```
products <- c("Product A", "Product B", "Product C",
              "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)

barplot(quantity,
        names.arg = products,
        xlab = "Product Name",
        ylab = "Quantity Available",
        main = "Product Inventory Quantity")
```

OUTPUT SCREENSHOT



RESULT

Thus, the inventory quantity bar chart was created successfully in RStudio. Product C has the highest available quantity.

QUESTION 2 – STACKED BAR CHART: PRODUCT QUANTITY BY CATEGORY

AIM

To create a stacked bar chart in RStudio showing product quantities within different product categories.

ALGORITHM

1. Create product, category and quantity vectors.

2. Create an inventory data frame.

3. Use xtabs() to create a product-category table.

4. Use barplot() to generate stacked bars.

5. Add labels, title and legend.

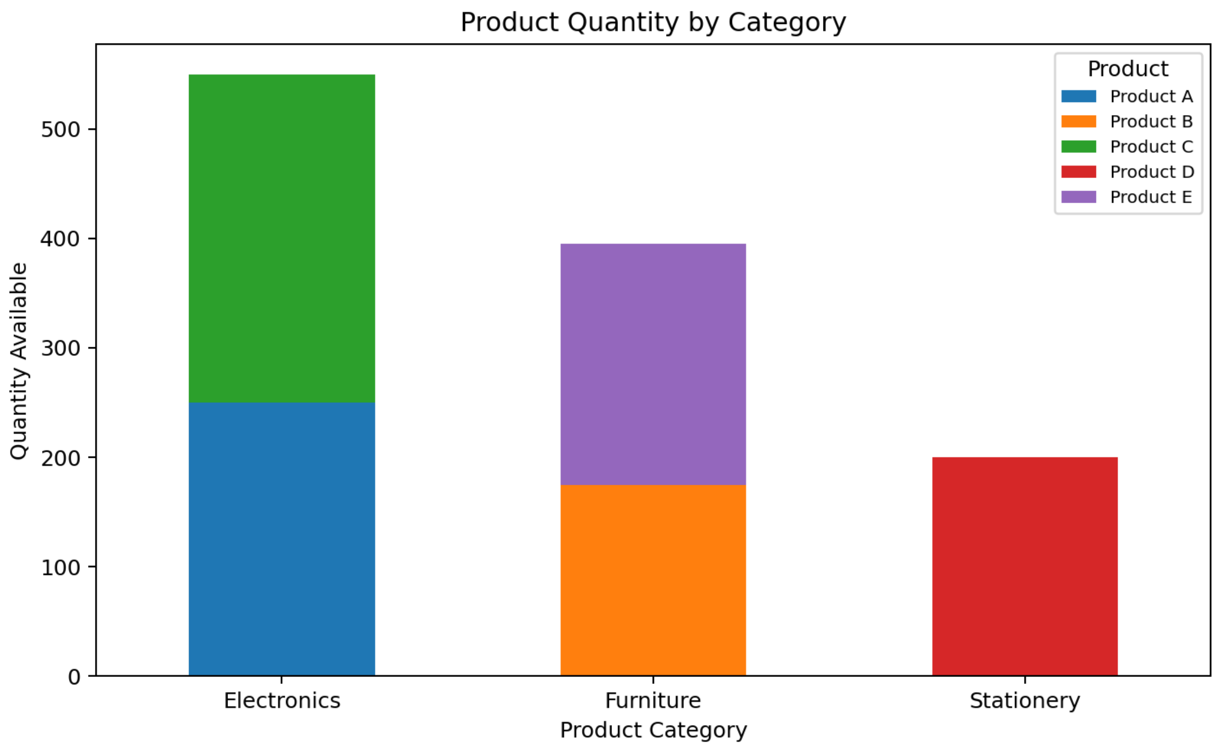
PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

```
product <- c("Product A", "Product B", "Product C",
             "Product D", "Product E")

category <- c("Electronics", "Furniture", "Electronics",
              "Stationery", "Furniture")

quantity <- c(250, 175, 300, 200, 220)

inventory <- data.frame(product, category, quantity)

quantity_table <- xtabs(quantity ~ product + category,
                        data = inventory)

barplot(quantity_table,
        beside = FALSE,
        xlab = "Product Category",
        ylab = "Quantity Available",
        main = "Product Quantity by Category",
        legend.text = rownames(quantity_table))
```

OUTPUT SCREENSHOT



RESULT

Thus, the product quantity distribution by sample category was displayed successfully using a stacked bar chart in RStudio.

QUESTION 3 – SCATTER PLOT: PRODUCT PRICE VS QUANTITY

AIM

To create a scatter plot in RStudio and explore the relationship between product price and quantity available.

ALGORITHM

1. Create sample price and quantity vectors.

2. Use plot() to create the scatter plot.

3. Add labels, title and grid.

4. Calculate correlation using cor().

5. Interpret the relationship.

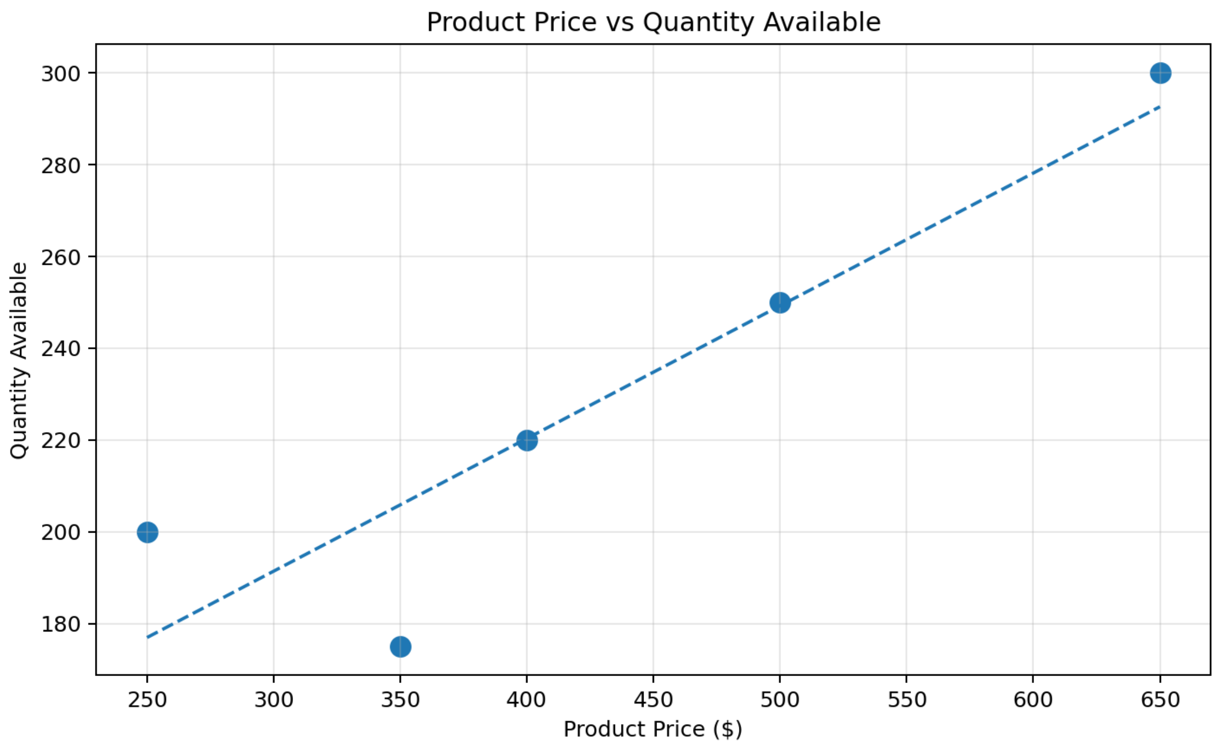
PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

```
product_price <- c(500, 350, 650, 250, 400)
quantity_available <- c(250, 175, 300, 200, 220)

plot(product_price, quantity_available,
     pch = 16,
     xlab = "Product Price ($)",
     ylab = "Quantity Available",
     main = "Product Price vs Quantity Available")

grid()

correlation <- cor(product_price, quantity_available)
print(correlation)
```

OUTPUT SCREENSHOT



RESULT

Thus, the scatter plot was created successfully in RStudio. The sample values suggest a generally positive relationship between product price and quantity available.

QUESTION 4 – TABLEAU DASHBOARD: PRODUCT INVENTORY

AIM

To prepare the product inventory data in RStudio and develop a Tableau dashboard containing the bar chart and stacked bar chart.

ALGORITHM

1. Create the product inventory data frame in RStudio.

2. Export the data as CSV.

3. Connect Tableau to the exported file.

4. Create the product quantity bar chart.

5. Create the category stacked bar chart.

6. Combine both worksheets in a Tableau dashboard.

PROCEDURE IN RSTUDIO

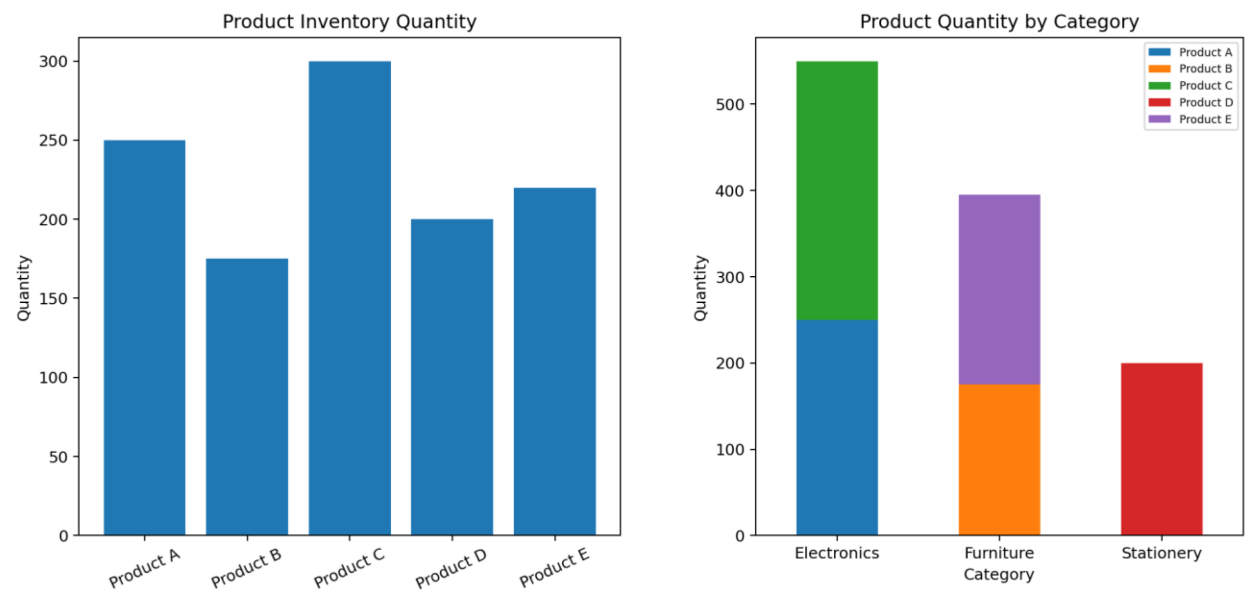
Open RStudio and execute the R code below. The Product_Inventory.csv file is saved in the current working directory. Run getwd() to find the folder. Open Tableau and connect to the CSV file. Create the required worksheets and combine them in a dashboard.

R PROGRAMMING CODE FOR RSTUDIO

```
inventory <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C",
                   "Product D", "Product E"),
  Category = c("Electronics", "Furniture", "Electronics",
               "Stationery", "Furniture"),
  Quantity_Available = c(250, 175, 300, 200, 220)
)

write.csv(inventory,
          "Product_Inventory.csv",
          row.names = FALSE)

getwd()

# TABLEAU STEPS:
# Sheet 1:
# Product_Name -> Columns
# SUM(Quantity_Available) -> Rows
# Marks -> Bar
#
# Sheet 2:
# Category -> Columns
# SUM(Quantity_Available) -> Rows
# Product_Name -> Color
#
# New Dashboard -> Add Sheet 1 and Sheet 2
```

OUTPUT SCREENSHOT

Product Inventory Dashboard – Tableau View



Dashboard output preview: inventory bar chart and category stacked bar chart combined

RESULT

Thus, the product inventory data was prepared in RStudio and exported successfully for creating the interactive Tableau dashboard.